# Kaya

## A multi-organ digital twin for Type 2 Diabetes

It shows what glucose does to the liver, eyes and kidneys, and lets a doctor test a change first.

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# The damage from diabetes is silent

Glucose is tracked. The organs it harms are checked once a year, if at all.

**100M+**

people with diabetes in India (ICMR-INDIAB)

**21.7%**

of Indians with diabetes had retinopathy (AIOS screening)

**22%**

of US adults with diabetes have advanced liver stiffness (NHANES)

- Fatty liver, retinopathy and kidney disease build for years before symptoms.
- Existing twins and apps forecast sugar and coach diet. None we reviewed links it to organ damage.

# What exists, and the gap

Reviewed from public sources

| Who | What it does | Gap |
|---|---|---|
| **Twin Health** | Remission program: CGM, wearables, care team | Reports liver and weight benefit; closed program |
| **Fitterfly, sugar.fit, Ultrahuman** | CGM plus coaching and scores for Indian users | Glucose and lifestyle; no organ outcomes |
| **Dexcom, Roche SmartGuide** | Predictive low-glucose alerts 20-30 min ahead | Glucose only |
| **Remidio + RetinaRisk** | Offline smartphone DR screening, risk-based interval | Eye only; no wearable link |
| **ARTEMIs, HEPATWIN (EU)** | Virtual organ twins for liver disease | Research stage; liver only |
| **Kaya** | One twin: glucose to liver and eye, plus what-if | **Open; validated per module** |

# What Kaya does

Two data streams fused into one twin, four questions answered, one place to test a change

Static: labs, genetics, history + Dynamic: CGM, heart rate, sleep, steps, meals

### Glucose

What will glucose do in the next 2 hours? Is an event coming?

### Liver

Is fat building up? Is fibrosis developing?

### Eye

What is the retinopathy grade and where is it heading?

### Kidney (next)

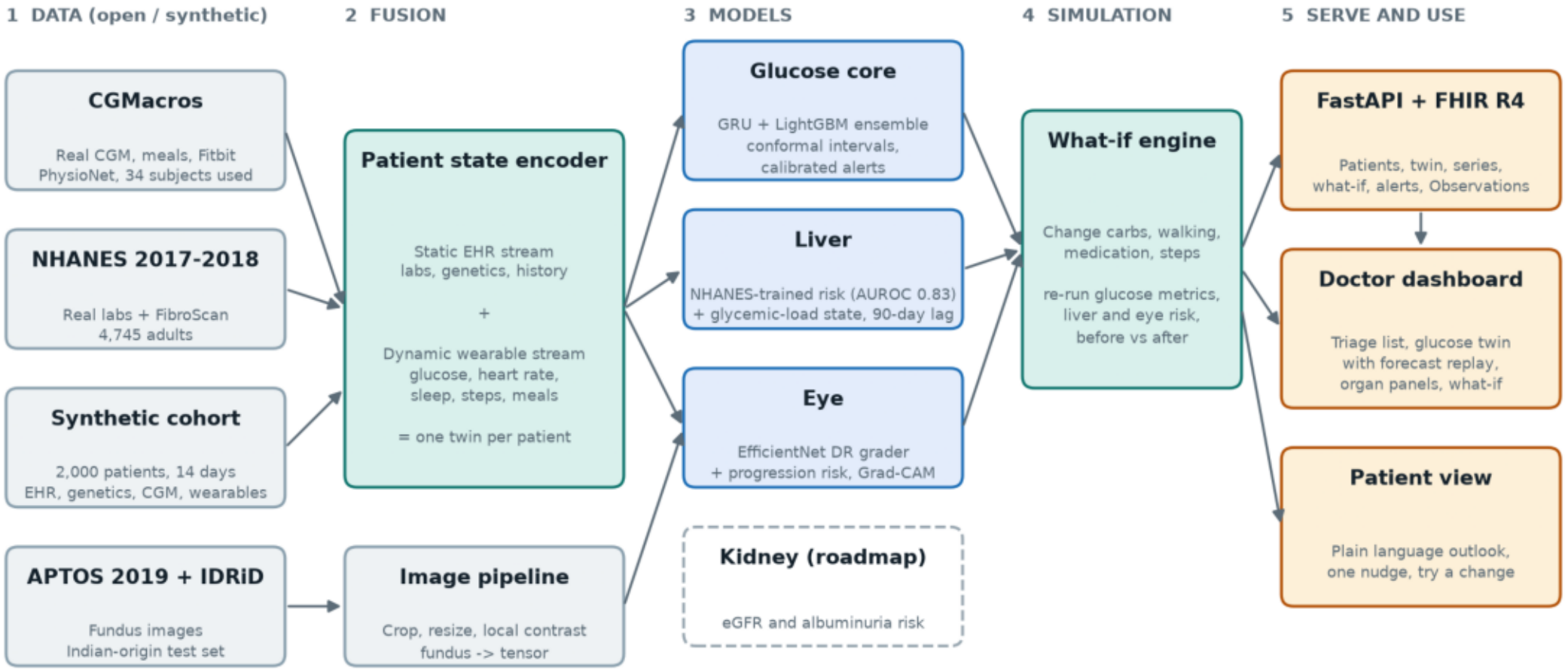
Is kidney function declining?

### What-if engine

Change carbohydrates, walking, medication or steps, and see the glucose curve and one-year organ risk move

# Kaya: system architecture

Multi-organ digital twin for Type 2 Diabetes. Open and synthetic data only.




Validation: patient-level splits, real-data checks per module, calibration, ablations | Governance: synthetic or open data only (DPDP-aware), decision support not diagnosis, model cards

# Data: open or synthetic only

What is real, what is simulated, and how we validate

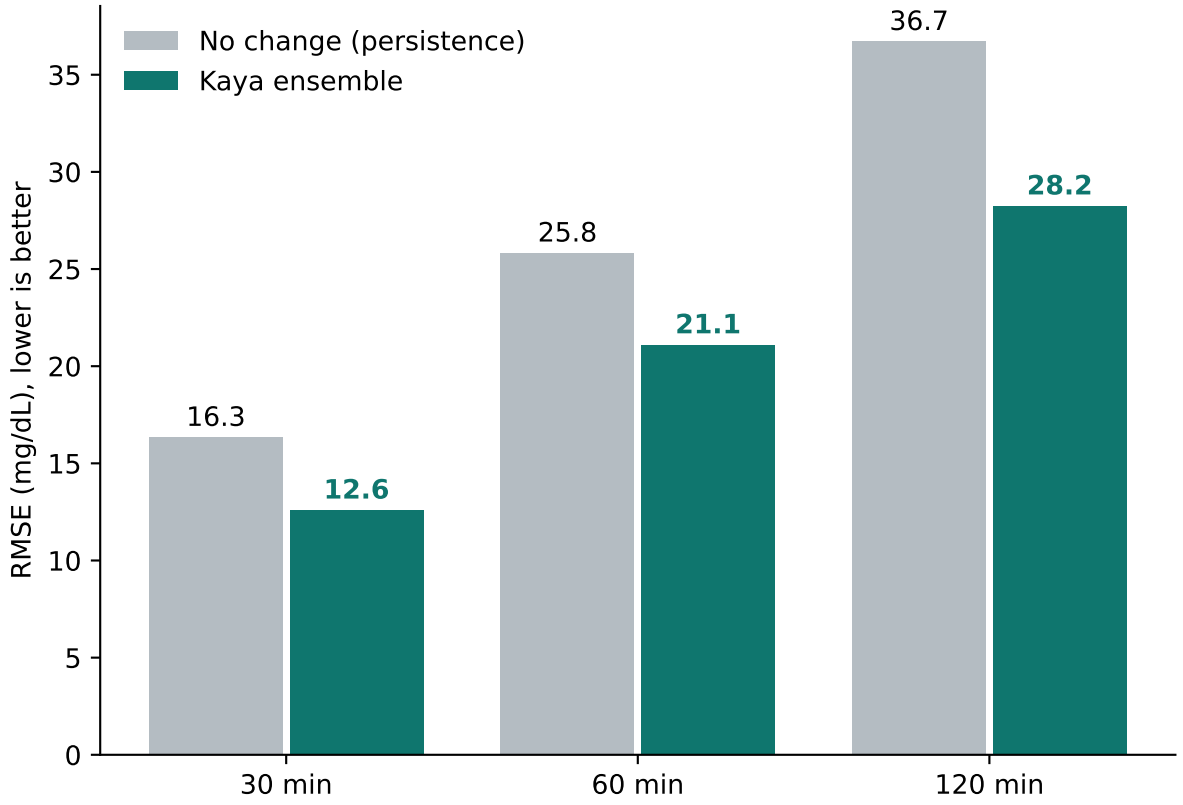
| | | |
|---|---|---|
| **CGMacros (PhysioNet)** | **Real** | Glucose forecaster: 34 subjects, subject-level cross-validation |
| **NHANES 2017-2018** | **Real** | Liver model: 4,745 adults with FibroScan labels |
| **APTOS 2019 + IDRiD** | **Real** | Retina grader: train on APTOS, test on Indian IDRiD |
| **Synthetic cohort** | **Simulated** | 2,000 patients: links glucose patterns to organs for the twin and what-if |

**Why this is honest**

No public dataset links one person's glucose to liver, eye and kidney outcomes. Each module is checked on real data where it exists.
The glucose-to-organ links and what-if effect sizes are editable, literature-informed assumptions, labelled as simulation in the product.

# Glucose forecasting on real, held-out subjects

CGMacros, 34 subjects, subject-level 5-fold cross-validation




**72%**

of low-glucose episodes caught

**45 min**

median warning before the low

**80%**

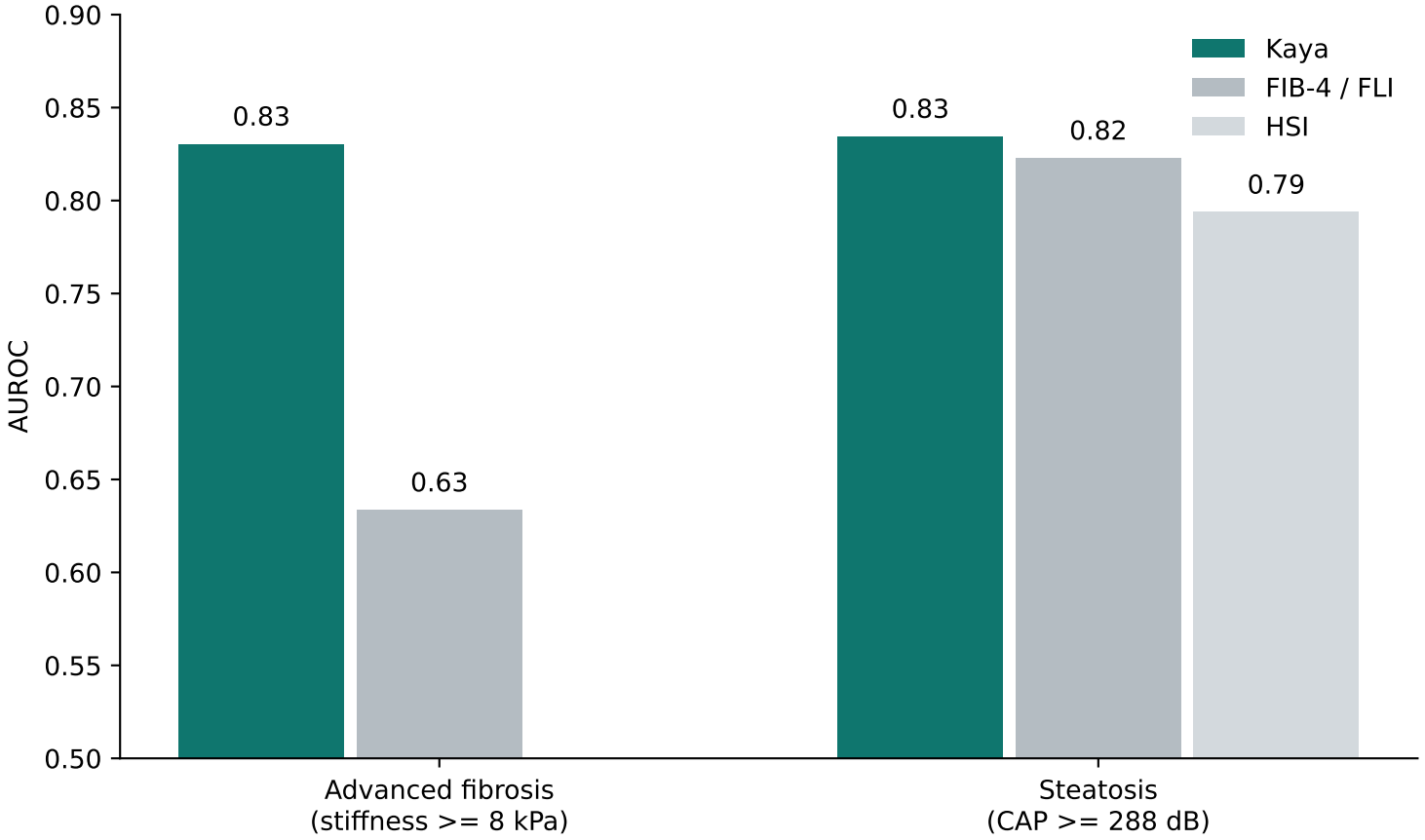
interval coverage (conformal, exact)

**~1/day**

false alerts per patient at that operating point

# Liver: validated on real FibroScan data

NHANES 2017-2018, 4,745 adults, 5-fold cross-validation, 95% CIs in the repository




- Fibrosis: 0.83 vs 0.63 for FIB-4
- Steatosis: 0.83 vs FLI 0.82, HSI 0.79
- Holds in adults with diabetes (0.76 / 0.80) and in A
- On top: a transparent glycemic-load state that mo

# Eye: retinopathy grading, tested on Indian images

EfficientNet-B2 trained on APTOS 2019, tested on IDRiD (Indian), ordinal grading

**0.89**

quadratic kappa
APTOS held-out

**0.98**

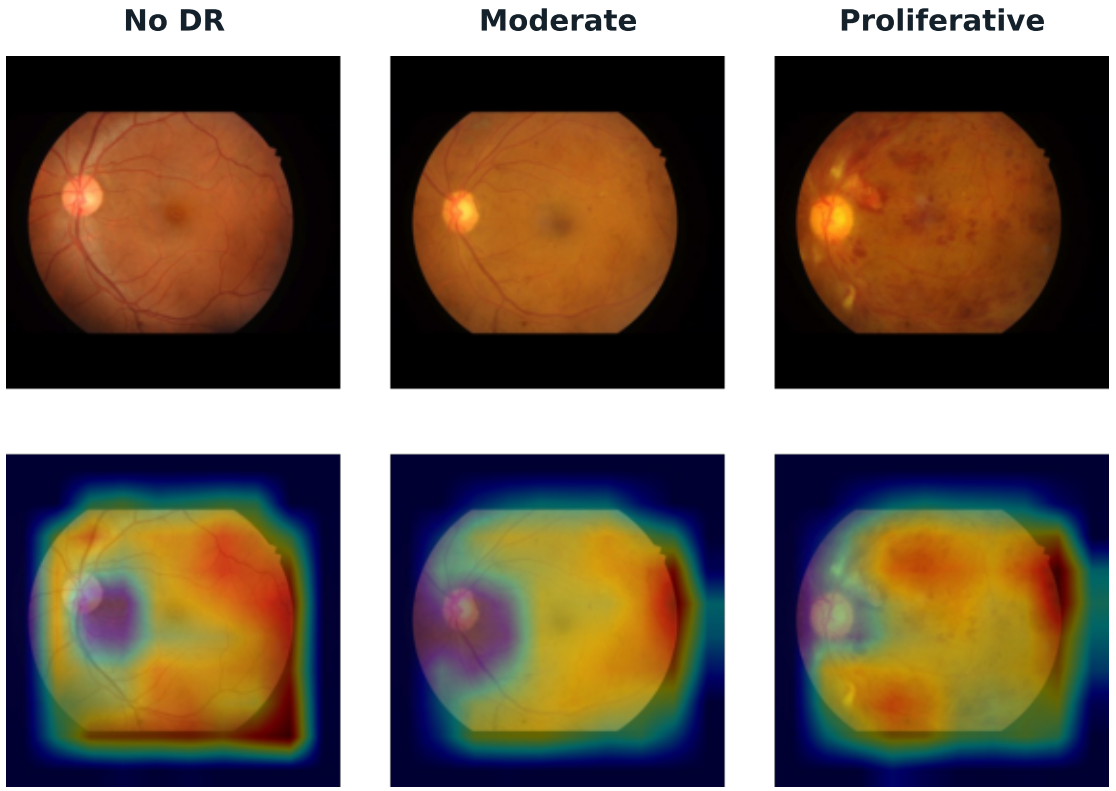
referable DR AUROC
APTOS held-out

**0.71**

quadratic kappa, IDRiD test
(Indian, fine-tuned locally)

**0.92**

referable DR AUROC
IDRiD test




Top: IDRiD fundus.  Bottom: Grad-CAM, where the model looked.

Without fine-tuning on IDRiD: kappa 0.57, AUROC 0.88. Camera and population shift is real, and local data closes part of the gap.

# What-if: test the change on the virtual patient

Patient 1705, last 14 days replayed with three changes together

| | Today | With the change |
|---|---|---|
| Time in range | 42% | **71%** |
| Time above 180 | 58% | **29%** |
| Liver steatosis risk, 12 months | 26% | **11%** |
| Advanced fibrosis, 12 months | 35.5% | **24.4%** |
| Retinopathy progression, 2 years | 6% | **2%** |

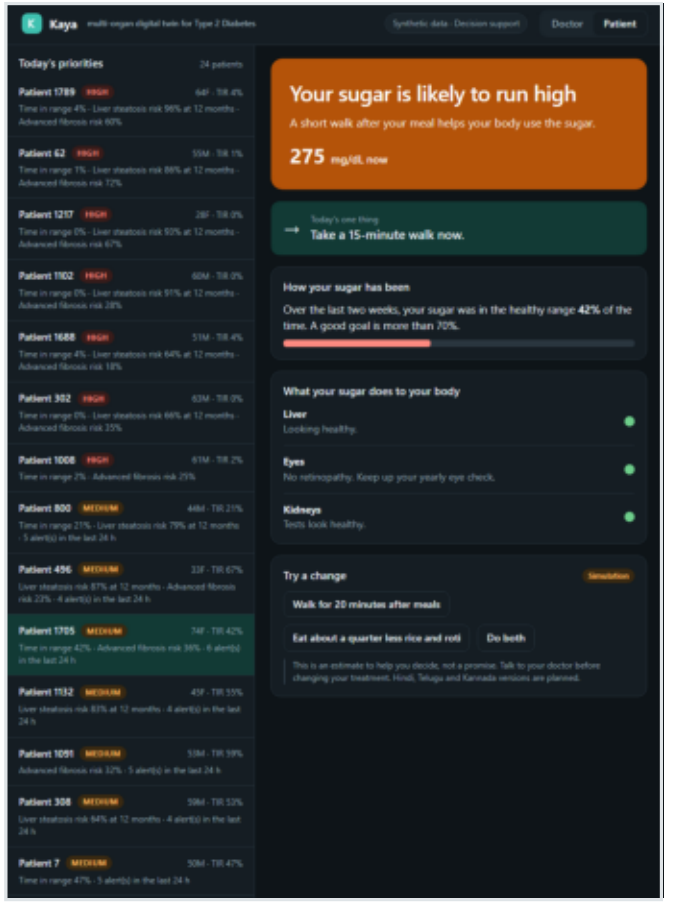
Walk 20 min after meals, 25% smaller carbohydrate portions, medication optimised. Effect sizes are editable, literature-informed assumptions and shown to the clinician as a simulation, not a prediction.

# The product

Doctor workspace and patient view. Replay shows the forecast against what actually happened.




Doctor view




Patient view

# Limitations, ethics and what is next

## Limitations we state openly

- Glucose-to-organ links and what-if effect sizes are assumptions, not clinical findings
- Glucose model validated on 34 real subjects; patients in the demo are synthetic
- Decision support for clinicians, not a diagnostic device; not clinically validated

## Roadmap

- Kidney risk module and an illustrative, clearly labelled retina progression preview
- Hindi, Telugu and Kannada patient view; PDF clinical summary
- Consent records, audit log, role-based access, fuller FHIR and ABDM alignment
- Validate the glucose-to-organ links on real longitudinal data when available

# Existing twins manage sugar.

Kaya shows what sugar does to the organs, and lets a doctor test the fix first.

Open source, Apache-2.0 · open and synthetic data only · github: see README

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